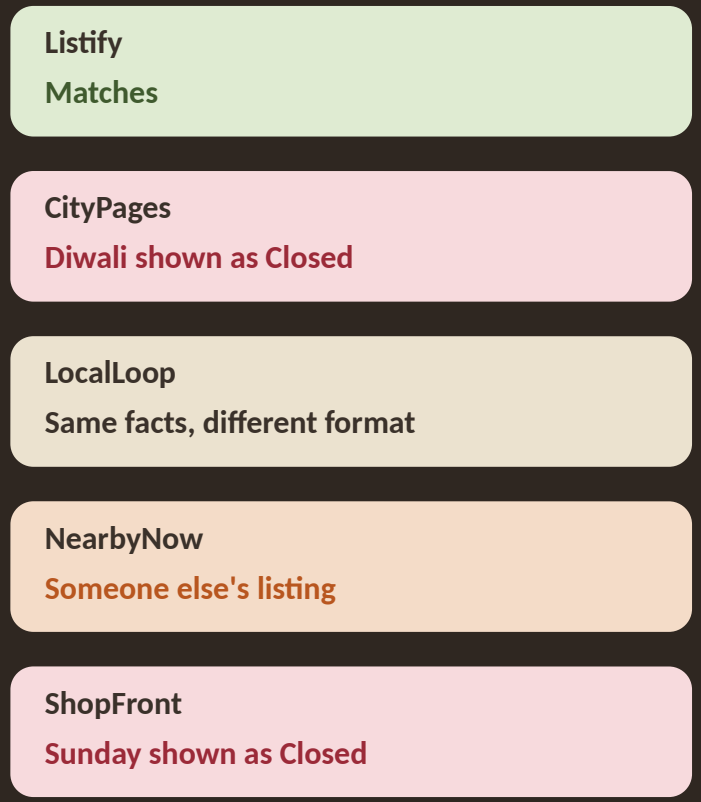
AGENTIC AI HACKATHON 2026 · BUSINESS & PRODUCTIVITY AI

Business Listing Consistency Agent

One true record. Five listing sites. An agent that finds every mismatch, fixes only what you approve, and proves the fix went live.



Live demo zirohacksub.onrender.com **demo PIN** 7KQ4X-M9PHT **Repo** github.com/yourgrok/zirohacksub

Fictional business (Sunrise Sweets & Bakery, Pune), mock listing sites, synthetic data only.

OVERVIEW

The seven things you asked us to show

1

The problem

Five sites, five versions of one business.

slides 3-4

2

Our solution

Scan, review, apply, verify. The owner stays in charge.

slides 5-6

3

The agentic AI

A loop that reads, decides, acts, verifies and recovers.

slides 7-8

4

Technical implementation

Architecture, stack, data flow, platform integration, guardrails.

slides 9-11

5

The innovation

Why it is not a dashboard, a chatbot or a cron job.

slide 12

6

The demonstration

One scenario end to end, the result on screen, and the TC1-TC8 proof.

slides 13-15

7

The impact

Numbers we measured, scalability, honest limits, roadmap.

slides 16-17

TRY IT YOURSELF

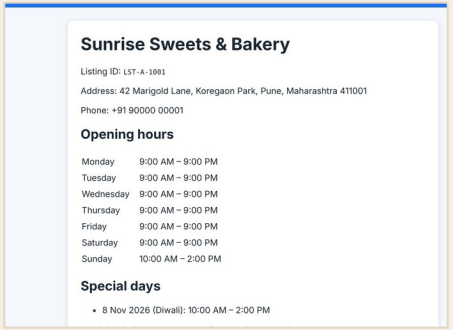
zirohacksub.onrender.com

PIN 7KQ4X-M9PHT

2-minute video: docs/demo/demo.mp4

One bakery, five listing sites, five different stories

A · Listify



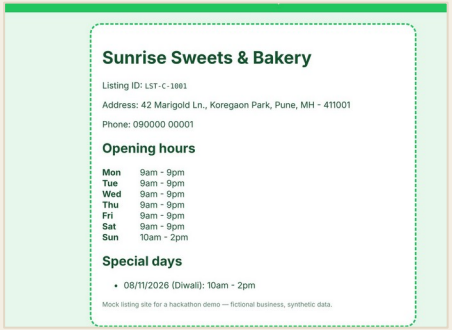
Matches the record

B · CityPages



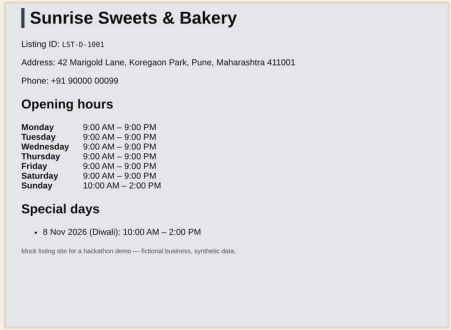
Diwali shown as Closed (record: 10:00-2:00)

C · LocalLoop



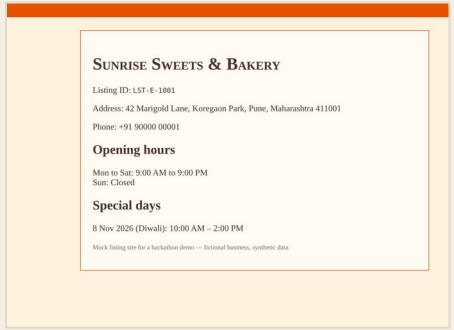
Phone and address written differently, same meaning

D · NearbyNow



Other phone number; claimed by another account

E · ShopFront



Sunday shown as Closed (record: 10:00-2:00)

What is a listing inconsistency?

Any difference between what the owner means and what a listing site shows: name, address, phone or hours. Some are obvious (a wrong holiday). Some only a careful reader catches, and some look different but mean the same thing.

In our test bed (fictional bakery, Pune)

- **Wrong holiday hours** on one site (B)
- **Same facts, different format** (C): "090000 00001" equals "+91 90000 00001"
- **Stale phone** on a listing that another account has claimed (D)
- **Regular hours differ** (E): Sunday listed as Closed

Why it matters, and why owners can't fix it by hand

WHY IT MATTERS

Customers are turned away

A holiday or Sunday shown as Closed sends paying customers elsewhere. The reverse sends them to a locked door.

Calls go to the wrong number

A stale phone number on one site means missed orders and a business that looks careless.

Conflicting details cost trust

Every platform shows its own version of the business, and customers notice when versions disagree.

WHY OWNERS STRUGGLE

Many sites, many formats

Each site stores phones and hours its own way. We parse four different hours layouts.

Every change is repeated

One new holiday means one edit per site, each with its own login and form.

Not every listing is yours

A listing claimed by someone else cannot be edited. A bot must not try.

Edits do not always go live

Moderation queues and dropped edits: "saved" is not the same as "live".

Look-alike businesses

"Sunrise Sweet House" sits next to our bakery. Editing the wrong one is worse than doing nothing.

Nobody checks afterwards

Drift returns silently, so the cleanup never stays done.

By hand it is slow and gets forgotten. Letting a bot edit freely is unsafe. The real problem is trust.

Tell us the truth once. The agent keeps every site honest.

1 — Scan

Reads all five sites and compares meaning, not formatting: 25 field checks in about 150 ms.

2 — Review

Every proposed fix is a card with a plain-English reason. You approve or reject each one.

3 — Apply

Only approved changes are sent, and never to a listing you do not own.

4 — Verify

The agent reloads the public page and marks VERIFIED only if the value is really there.



Approvals: what the site shows versus what your record says, and why.

Your record is the single source of truth

Name · Address · Phone · Weekly hours · Holidays

Five kinds of fact, one place. Nothing reaches a listing until you approve it.

Who it helps, and what changes for them

WHO IT HELPS

Busy owners of local shops

Bakeries, clinics, salons: no time to check five sites, no SEO team.

Owners with more than one business

Each business has its own PIN and fully separate data.

New owners

The agent finds where you are already listed and adds you where you are missing, after you approve.

	Today, by hand	With the agent
Find mismatches	Open each site and compare by eye	One scan reads every site and every field
Same meaning, other format	Easy to miss, or to "fix" needlessly	Compared by meaning; formatting-only differences are ignored
Fix them	Log in to each site and retype	Review cards and approve; only approved changes are sent
Know it worked	Hope the site saved it	Reloads the live page; moderation shows as pending
Owner effort in our demo	Edit 2 sites by hand, re-check all 5	3 clicks: Scan, Approve all, Apply
Stay consistent	A one-off cleanup	Repeatable scans and holiday alerts

A goal-driven loop, not a chat box

Goal every listing says what the owner's record says. The owner approves; the agent does the legwork and proves the result.

1 Perceive

Reads each public page and owner dashboard. Raw values are kept.

`read_listing, normalise`

2 Decide

Compares meaning per field and picks the next tool from the current state.

`compare, planner`

3 Act

Proposes a fix, flags a listing, asks the owner, or applies what was approved.

`propose, flag_listing, apply_approved`

4 Verify

Reloads the page. VERIFIED only if the approved value is really there.

`reload_and_verify`

5 Recover

Page changed? Site down? Edit dropped? Re-ask, retry, or re-send once.

`same tools, new decision`

loops until nothing is left to do

EIGHT TOOLS, ONE REGISTRY

read_listing page + dashboard	**normalise** E.164, address, hours	**compare** meaning per field	**propose** one card per fix
apply_approved approved rows only	**reload_and_verify** checks the live page	**flag_listing** not ours: no write	**ask_owner** asks for approval

Who decides each step?

A rules planner (default, no key needed) or an optional Gemini planner. Same tools, same guardrails. Each trace row carries a badge: Gemini, Deterministic, Fallback or Blocked.

From plain-language request to verified result

Interprets requests

- Typed text becomes an intent: a question, a detail edit or a closure date.
- chrono-node plus rules turn "26th December" into Sat 26 Dec 2026 (IST).
- Unclear input gets exactly one clarifying question.
- Optional Gemini cross-checks the reading; if it disagrees, the owner chooses.
- Nothing is saved before Confirm.

"closed on 26th December" > Action: Closed, Date: Sat 26 Dec 2026

Selects tools

- Each step the planner sees the tool list, the record and every site's state.
- Nothing read yet: read_listing. Then normalise, then compare.
- Owner's listing differs: propose. Permission denied: flag_listing (forced in code).
- Approved rows: apply_approved.
- An unsafe pick is blocked and the rules choose instead.

Rogue planner test: 6 unsafe picks, 6 blocked, 0 writes

Runs multi-step workflows

- Scan phase reads and proposes. Execute phase applies and verifies.
- One closure request fans out to every site (TC5: 3 verified, 1 flagged, 1 pending moderation).
- 27 tool calls for the TC1 holiday fix, each a visible trace row.
- Every run ends with a short "what I did and why" briefing.

TC1: MISMATCH > PROPOSED > APPROVED > APPLIED > VERIFIED

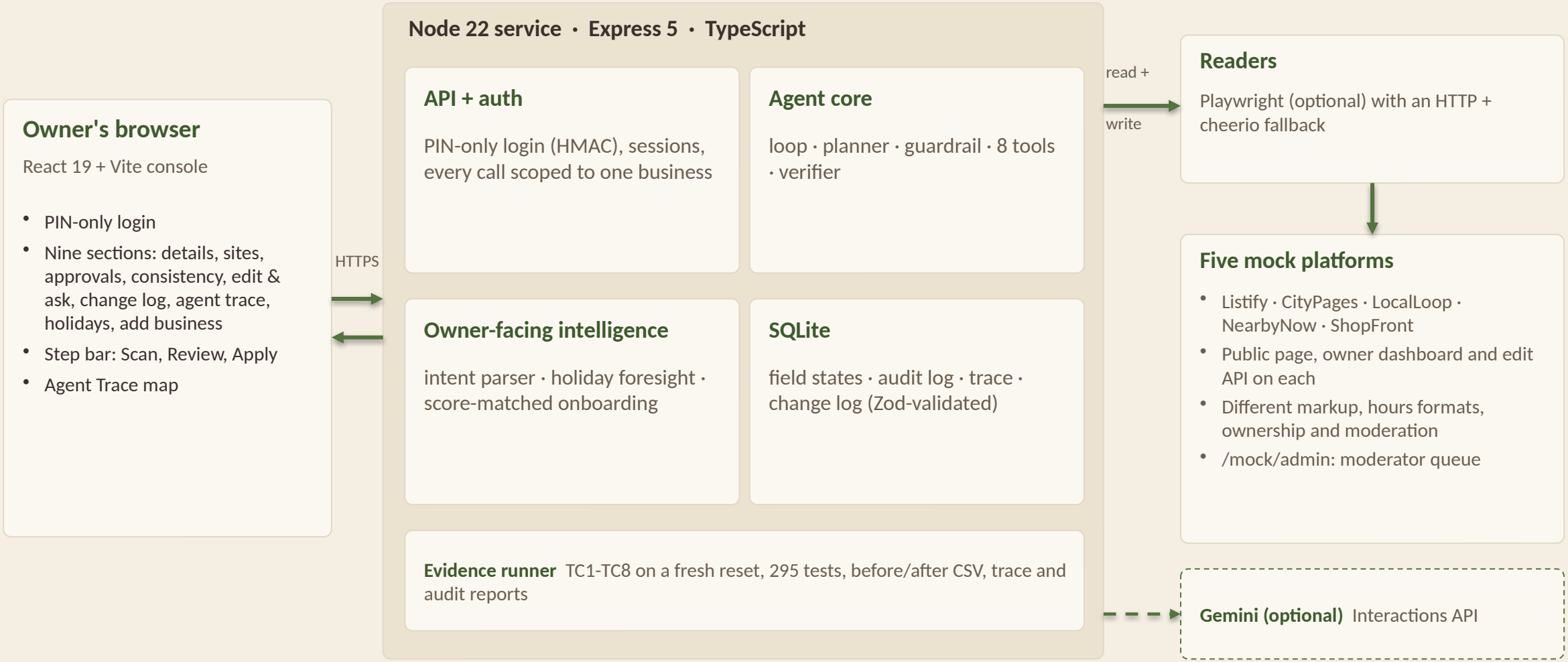
Verifies and recovers

- Reloads and compares. "Saved" is never trusted.
- Moderated site: PENDING_MODERATION until the page really changes (TC4).
- Page changed after approval: sends nothing, asks again with the new facts (TC7).
- Site down: retries. Edit dropped: re-sent once, then verified (TC8).

TC8: outage retried, lost write re-sent, VERIFIED

Honest note The default decision-maker is a rules planner. The Gemini planner is optional and falls back to rules on 429, timeout or invalid JSON; it is covered by 12 mocked-client tests, and no live Gemini run is in our evidence set (no key in the build environment).

One service: console, agent, readers, five mock platforms



Stack, data flow and how it talks to listing platforms

STACK

Language TypeScript end to end, Node 22

Server Express 5, Zod 4 validation

Console React 19 + Vite, served by Express

Data SQLite (better-sqlite3, node:sqlite fallback)

Readers Playwright, plus fetch + cheerio

Language tools chrono-node for dates

LLM (optional) @google/genai, Interactions API

Quality Vitest: 295 tests, 24 files

Deploy Docker, Render free tier, /health

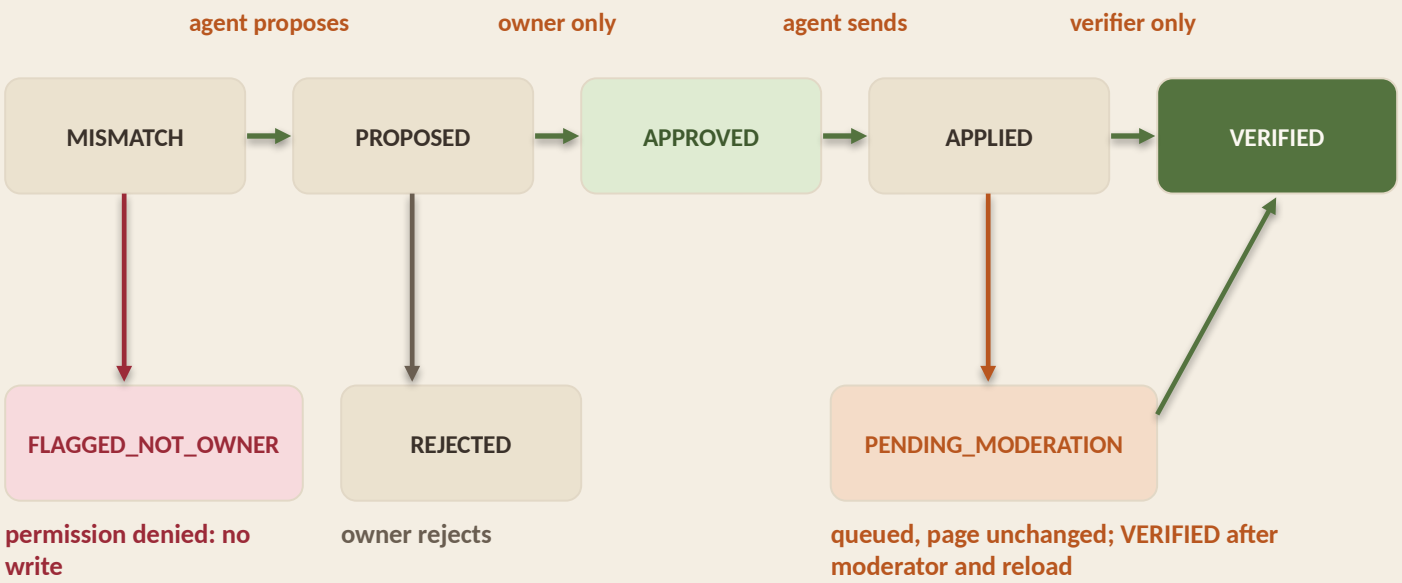
DATA FLOW OF A RUN

1. Owner record (SQLite) is the source of truth
2. Read each public page and dashboard; keep raw values
3. Normalise: E.164 phone, tidy address, parse hours
4. Compare meaning per field
5. Propose: one card per mismatch
6. Owner approves: DB row becomes APPROVED
7. Apply: read-before-write, idempotent
8. Reload and verify: VERIFIED, PENDING or MISMATCH
9. Report: trace, audit log, change log, CSV

INTEGRATING WITH PLATFORMS

- One reader per site turns four different hours layouts into one weekly schedule
- Every read asserts the page's listing ID equals the registered ID; look-alike listings are never read as ours
- Writes use each site's dashboard API with server-signed credentials; permission denied becomes a flag, never an edit
- Edit accepted but page unchanged and queued: PENDING_MODERATION
- Real platforms would need their own APIs and consent flows; we do not claim that yet

Safety is enforced in code, not in prompts



Entering APPROVED requires the actor "owner"; entering VERIFIED requires the actor "verifier". Enforced in src/core/states.ts and unit-tested.

Isolation and secrets

- One business can never read another's data: a session-scoped business ID is required on every call; foreign IDs answer 404.
- PINs are never stored (HMAC-SHA256, constant-time compare, rate-limited). The API key never reaches git, logs, /health or the trace.

GUARDRAILS IN CODE

- No apply without a DB row the owner approved
- Permission denied forces flag_listing
- Only the verifier can mark VERIFIED, after a reload
- Pending moderation is never reported as fixed
- Decoy listings are never read as ours or edited
- Re-running never double-applies

6 / 6

unsafe actions blocked by a scripted rogue planner: 0 writes, 0 false VERIFIED

Not a dashboard, not a chatbot, not a cron job

	Typical dashboard	Typical chatbot	Typical rule-based sync	Our agent
Compares meaning, not formatting	No	Partly	No	Yes
Knows who owns a listing and refuses to edit it	Partly	No	No	Yes
Reloads the page to prove the edit went live	No	No	No	Yes
Tells "pending moderation" from "done"	No	No	No	Yes
Explains every step it took (trace)	No	Partly	No	Yes
Recovers when a page changes or a site is down	No	No	Partly	Yes
Looks ahead (holiday alerts)	No	No	Partly	Yes

Compared with typical tools in each category, not with specific products. "Partly" means possible, or only if configured.

Verify-after-write

A field becomes VERIFIED only when the reloaded public page shows the approved value.

Ownership-aware

Claimed by someone else? No edit is attempted, and a claim request is drafted for the owner.

Finds you among look-alikes

Score = phone 40 + address 35 + name 25. The look-alike scores 37.7, the real listing 100.

Looks ahead, speaks plainly

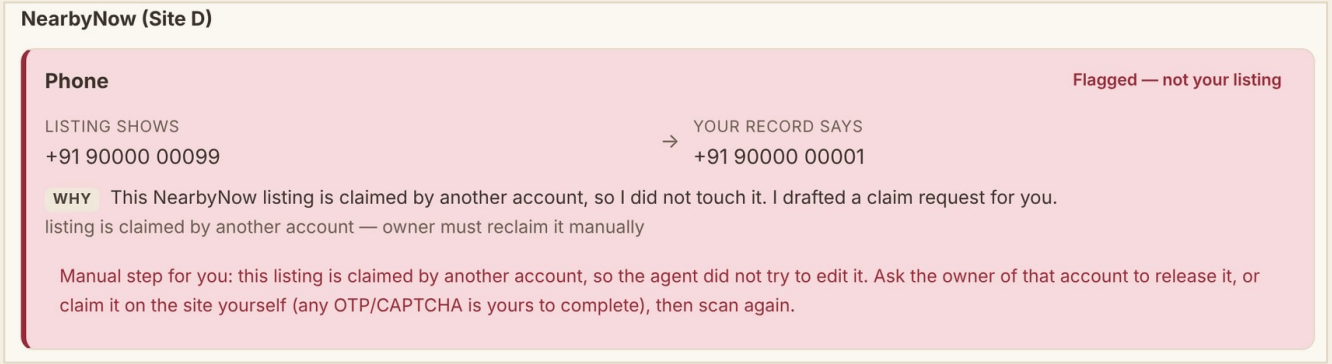
Holiday alerts at 30, 14 and 7 days. "Closed on 26th December" becomes a previewed, approved change.

End to end: holiday hours fixed, a stranger's listing left alone

1. **Scan** Five sites read in about 150 ms: 2 fixes proposed, 1 listing claimed by someone else, 2 format-only ignored.
2. **Review** CityPages lists Diwali as Closed; the record says 10:00-2:00. The card says why, and you decide.
3. **Safety** NearbyNow is claimed by another account: zero edits, and a claim request is drafted for you.
4. **Apply** Only approved changes are sent. Every step lands in the trace; each page is reloaded.
5. **Moderate** ShopFront queues the edit. It stays PENDING until its moderator approves; a re-check then verifies it.
6. **Report** Before/after table: 2 fixed and verified, 1 needs you, 2 ignored. CSV export.



Step 2, Review: the card shows both values and the reason.



Step 3, Safety: not our listing, so no edit is attempted and the manual steps are spelled out.

Watch the 2-minute video: docs/demo/demo.mp4 · Try it live: zirohacksub.onrender.com · demo PIN 7KQ4X-M9PHT

The result on screen: verified, pending or flagged

2 site(s) fixed and verified (CityPages, ShopFront); 1 need you to reclaim the listing (NearbyNow); 2 formatting-only difference(s) ignored.

All 25 | Needs attention 1 | **Verified 2** | Already fine 22 | All sites

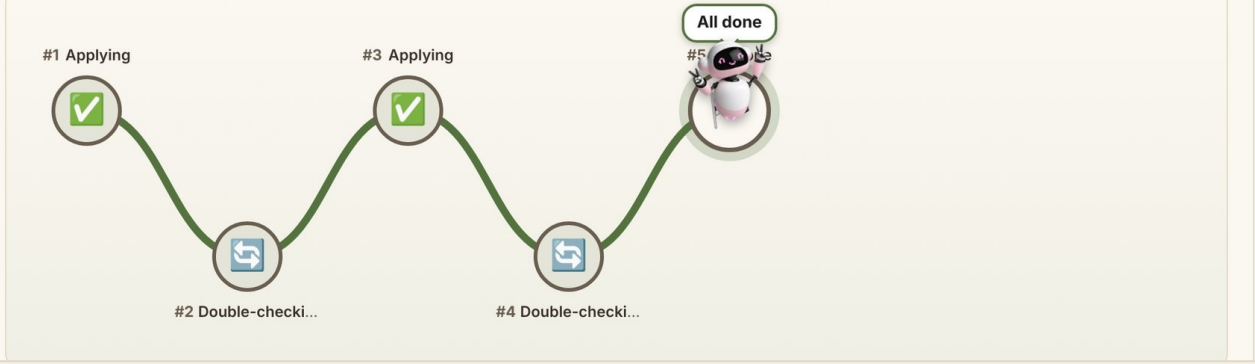
Site	Field	Original value	Intended value	Approval	Verified status
CityPages	Holiday 2026-11-08 (Diwali)	8 Nov 2026: Closed	8 Nov 2026: 10:00 AM – 2:00 ...	Approved	Verified
ShopFront	Regular hours	Mon to Sat: 9:00 AM to 9:00 P...	Mon-Sat 09:00-21:00; Sun 10:...	Approved	Verified

2 of 25 rows. Hover a cell to see the full text or the verification note.

Consistency table, filtered to Verified: both fixes were confirmed by reloading the page.

How to read it

- Verified: the reloaded page shows the approved value
- Pending: the site's moderator has not approved yet
- Flagged: not your listing, nothing was sent



Agent Trace: the route of the run, step by step.

Agent Trace: every step, in order

- Each row names the tool, who decided (Gemini, Deterministic, Fallback or Blocked), what it found and how long it took
- Verification is a visible step: Applying, then Double-checking, then All done
- Blocked actions show in red with the reason
- The little agent follows your pointer along the route

What the run proves: TC1 to TC8 pass from a fresh reset

Test case	Actual result
TC1 Holiday hours differ on one site	MISMATCH > PROPOSED > APPROVED > APPLIED > VERIFIED; 1 write
TC2 Phone formatted differently	FORMAT_ONLY_EQUIVALENT; 0 proposals, 0 writes
TC3 Listing claimed by someone else	FLAGGED_NOT_OWNER; 0 write attempts
TC4 Edit pending moderation	… PENDING_MODERATION > VERIFIED after moderator and reload
TC5 Plain-language closure	3 VERIFIED, 1 FLAGGED_NOT_OWNER, 1 PENDING_MODERATION
TC6 Foresight alert, no exception	"Bhai Dooj (11 Nov 2026) is in 21 days and has no opening-hours exception"
TC7 Listing changes after approval	Nothing sent; asked again with the new facts, then VERIFIED
TC8 Site down, then drops an edit	Outage retried; lost write re-sent once; VERIFIED

Deterministic run with the date pinned to 21 Oct 2026; each test starts from a fresh reset. The Gemini run needs an API key and is not part of this evidence.

8 / 8

official test cases pass from a fresh reset

295

automated tests pass across 24 files

6 / 6

unsafe actions blocked in the rogue-planner test, with 0 writes

Re-run it yourself open /tests, press Run all

Value we can measure today, and where it scales

about 150 ms

full scan: 5 sites x 5 fields = 25 checks (local mock sites; three runs: 139-156 ms)

3 clicks

owner effort in the demo: Scan, Approve all, Apply

0

unapproved writes, decoy edits or false VERIFIED in the tests

8 / 8

official test cases pass from a fresh reset

6 / 6

unsafe actions blocked in the rogue-planner test

295

automated tests pass

Practical value

- Hours saved on every holiday change, and no more wrong-hours surprises
- Safe by design: no write without approval, none to listings you do not own
- Owners get proof (VERIFIED, trace, CSV), not promises

Scalability

- Each new platform is one more reader; the mock sites already differ in markup, formats and rules
- Work per scan grows with sites x fields (25 checks today)
- Every business is isolated by a session-scoped ID; one Docker service on a free host

Future potential

- Real platform APIs and consent flows; OTP and phone checks stay with a human
- Scheduled scans and alerts when drift returns
- More fields (categories, photos, menus) and a multi-location view

What is real today, and what comes next

Real and tested today

- Scan, approve, apply, verify across five mock platforms
- Ownership flags, moderation handling, retries and re-planning (TC1-TC8)
- PIN-only login, per-business isolation, score-matched onboarding
- Plain-language edits and questions, holiday foresight
- Trace, change log and CSV evidence; 295 tests
- Deployed demo on a free host (first load can take 30-60 s)

Mocked or not built yet

- The platforms are mock sites; real ones need their own APIs and consent flows
- Claiming a listing is mocked (no OTP). On real sites that step stays human: the agent never automates OTP or CAPTCHA
- Festival dates must be confirmed by the owner (approximate calendar)
- No live Gemini run in the evidence (no key in the build environment)
- A lost PIN cannot be recovered; no phone verification yet

NEXT STEPS

1. **A real platform adapter with its consent flow**
2. **Scheduled scans and drift alerts**
3. **PIN recovery and phone verification as a human step**
4. **More fields and a multi-location view**

IN ONE SENTENCE

An agent you can trust near your public listings.

Safe by construction

No write without approval, none to listings you do not own.

Proven

TC1-TC8, 295 tests, plus trace and CSV evidence.

Ready to try

Live demo, a 2-minute video and the open repo.

Live demo zirohacksub.onrender.com **demo PIN** 7KQ4X-M9PHT (synthetic)

Video docs/demo/demo.mp4 **Repo** github.com/yourgrok/zirohacksub

Thank you. Questions?